Prosperity Prognosticator: A Machine Learning Approach to Startup Success Prediction

1. Introduction

Project Title: Prosperity Prognosticator: Machine Learning For Startup Success Prediction

Team ID: LTVIP2026TMIDS79953

Team Size:4

- **Team Leader**: Sumanth Suri - Project Lead & Model Developer
- **Team Member**: Narry Trisha - Data Engineer
- **Team Member:** Mb Tharun - Frontend & Backend Developer
- **Team Member**: Singana Sathish – Evaluation & Deployment Specialist

2. Project Overview

2.1. Purpose

Prosperity Prognosticator is a machine learning-based startup success prediction system designed to analyze business-related parameters such as funding amount, team experience, market size, revenue growth, and industry sector to predict the likelihood of startup success.

The main objective is to assist investors, entrepreneurs, and analysts in making data-driven decisions by leveraging predictive analytics and machine learning techniques.

2.2. Features

• Machine Learning Based Prediction
• Real-Time User Input Analysis
• Simple Web Interface using Flask
• Fast Prediction Response
• Expandable for Business API Integration

3. Architecture

3.1. Frontend

• Developed using HTML5 and CSS3
• Provides input form for startup parameters
• Displays prediction results dynamically using Flask

3.2. Backend

• Developed in Python using Flask Framework
• Loads trained model (model.pkl)
• Accepts user input
• Performs prediction using ML algorithm
• Returns results to frontend

Dataset:

• Funding Amount
• Number of Founders
• Years of Experience
• Market Size
• Revenue Growth Rate
• Industry Type

Model

Logistic Regression / Random Forest Classifier

Model Training Pipeline:

• Data preprocessing using Pandas
• Train-test split
• Model training using Scikit-learn
• Model evaluation (Accuracy, Precision, Recall)
• Model saved using Joblib (.pkl format)

3.3. Data Preprocessing:

• Handling missing values
• Encoding categorical variables
• Feature scaling
• Dataset splitting

3.4. Database

No database used in this prototype. Predictions are generated in real time.

4. Setup Instructions

4.1. Prerequisites

- Python 3.8+
- Flask
- Scikit-learn
- Pandas
- NumPy
- Joblib

4.2. Installation

```
pip install -r requirements.txt
python app.py
```

5. Folder Structure

5.1. Client (Flask Frontend)

```
templates/
├── index.html
├── chance.html
```

5.2. Server (Flask Backend)

App.py
Model.pkl
Startup_Model.ipynb
Requirements.txt

6. Running the Application

Python app.py
http://127.0.0.1:5000

7. API Documentation

This application works through form submission and internally uses:

Route: / **Method:** POST

http://127.0.0.1:5000/
http://127.0.0.1:5000/

Input:

- Funding Amount
- Founder Experience
- Market Size
- Revenue Growth
- Industry Type
- Startup Stage

Output:

Startup Success / Startup Failure prediction

8. Authentication

Not applicable – This is a prototype application.

9. User Interface

Two pages:

- index.html → Input page
- result.html → Result page

UI Highlights:

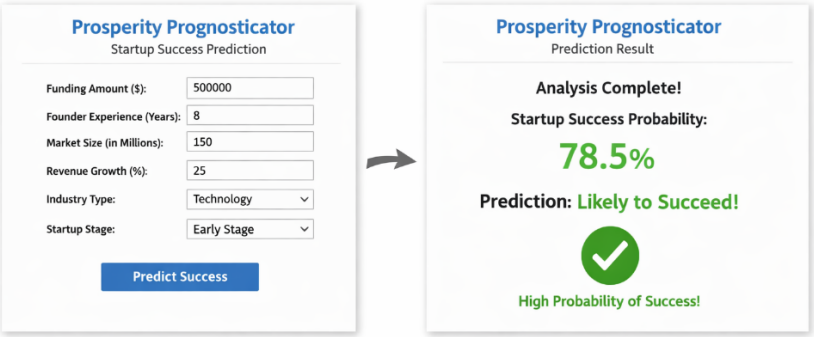
- Simple design
- Easy navigation
- Quick response
- Clear prediction display

10. Testing

- Accuracy evaluation using test dataset
- Manual testing with multiple inputs
- Confusion matrix generated

11. Screenshots

• User enters startup details
• System predicts success probability
• Result displayed on screen



12. Known Issues

• Accuracy depends on dataset quality

• No live startup funding API integration

• Limited dataset size

• Does not account for sudden market disruptions

13. Future Enhancements

• Integration with Startup & Investment APIs
• Deep Learning models for improved accuracy
• Mobile application
• Cloud deployment
• Investor analytics dashboard
• Real-time funding trend analysis